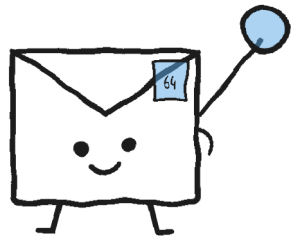
the little

PACKET

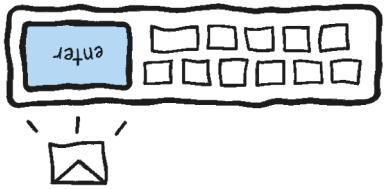
who wanted to see the world



a bedtime story for geeks of all ages

once upon a time...

someone typed **cats** into a search box and pressed enter.



and a little packet was born. it had a header, a payload, and a time-to-live of **64** hops.

it wanted to see the world.

first stop: DNS

"excuse me," said the packet. "where does search.example live?"



the wise old resolver checked her notes. "ah. **203.0.113.7**. off you go, dear."

routers. so many routers.

each one squinted at the address, pointed vaguely east, and said *"that way."*

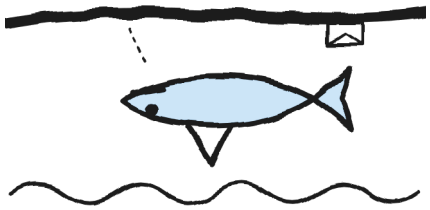
TTL 63 … 62 … 61 …

60 …

(the packet tried not to think about what happens at zero.)

under the sea

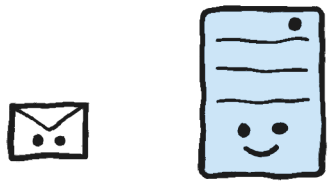
the packet rode a cable along the bottom of the ocean, as fast as light in glass.



a shark swam over for a nibble. (sharks really do this.)

knock knock

at last, the server! the packet knocked politely.



packet: **SYN?**

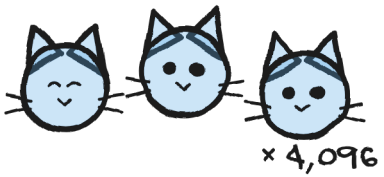
server: **SYN-ACK!**

packet: **ACK.**

and just like that, they were friends.

the payload

the server opened the packet's message. it said: **cats**.



it thought about this for three whole milliseconds, then sent back four thousand packets of cats.

(servers are like that.)

the end?



somewhere along the way, one little packet got lost.

don't worry. it'll get sent again. *it always does.*

tiny glossary
DNS — the internet's phone book
router — a very confident signpost
TTL — hops left before giving up
ACK — "got it!"

cut along the dashes